Revelink

Keep the curiosity.
Keep the learning connected.

Watch. Discover. Connect the dots.

Problem: learners lose context and momentum moving between video discovery, explanations, and saved learning.

TEAM SSRD **TEAM LEADER** Subram Das

A question should start a learning journey.

For adult students and early-career professionals in India, starting with English-language discovery.

THE STARTING MOMENT

“How do GPUs work?”

A learner finds a video, but the next question sends them to another tool.

THE FRICTION

Context gets lost.

Search, watch, ask, and remember are separate tasks. Unrelated feeds interrupt the journey.

THE IDEA

One connected workspace.

YouTube discovery + a Gemini teacher + organised topics + saved playback.

The learning context travels with the learner.

USP: video discovery, in-context explanation, and the next useful action in one continuous journey.

LEARNER NEED	FRAGMENTED WORKFLOW	REVELINK WORKFLOW
Find a useful starting point	Search for videos separately	Educational intent checks and learning-angle choices
Resolve the doubt	Explain the context again in chat	Teacher uses question, conversation and eligible video
Act on the explanation	Find the section or search again	Jump to an AI-matched section or open a real result
Return to the interest	Reconstruct the previous session	Organised subjects and saved playback position

Starting opportunity: adult self-directed learners exploring a concept, not a full course.

Built around discovery, clarity, and continuity.

01 / DISCOVER

Find your starting point

Search, video results, long videos and a short-video feed.

02 / UNDERSTAND

Ask beside the video

Follow-ups, simpler explanations and relevant video sections.

03 / CONTINUE

Resume your learning

Saved topics, watch history and last playback position.

04 / ORGANISE

Explore three Universes

Science & Technology; Commerce & Business; Arts & Humanities.

05 / STAY ON TASK

Guide learning intent

Clarify ambiguous intent; filter results using metadata.

06 / RESET

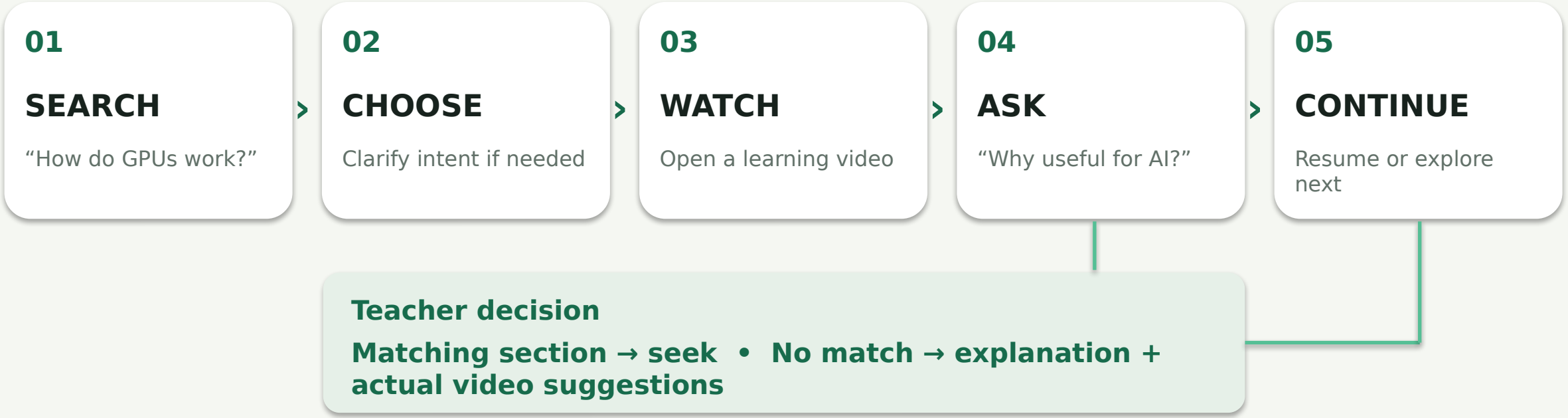
Take a bounded break

Optional topic-related clips after 12 active minutes; at most three.

Latest local build shown. Public demo is an earlier revision; learning outcomes have not yet been measured.

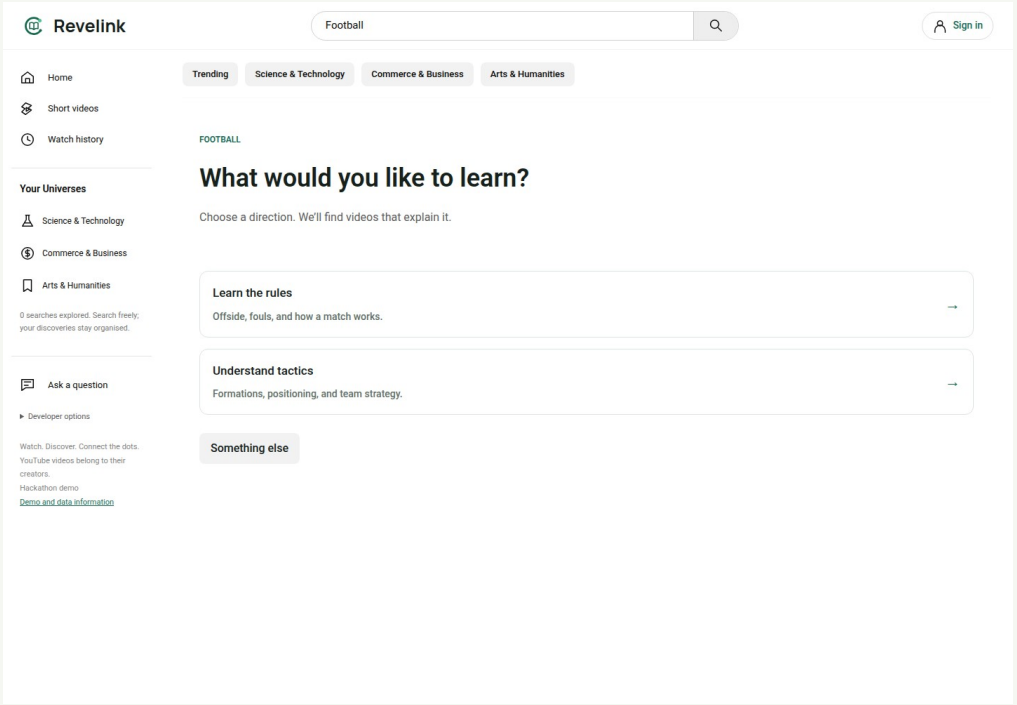
From “I wonder…” to “I can keep going.”

Illustrative GPU journey • implemented paths; AI decisions can be uncertain.

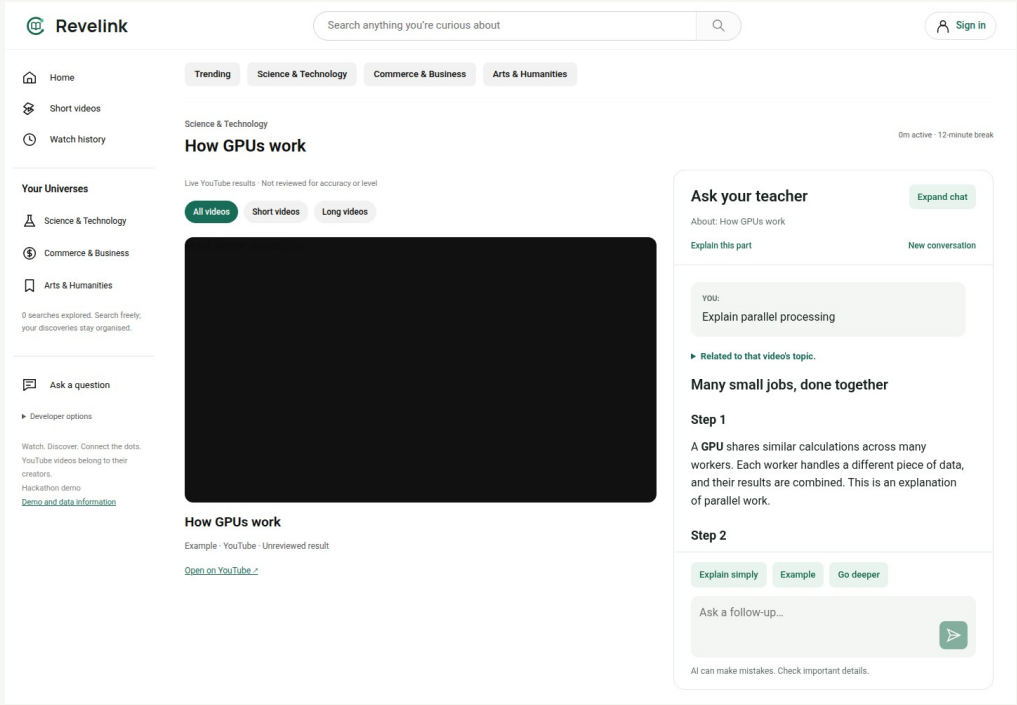


An interface designed around the learning task.

Actual local React interface • simulated API responses and player in these captures.



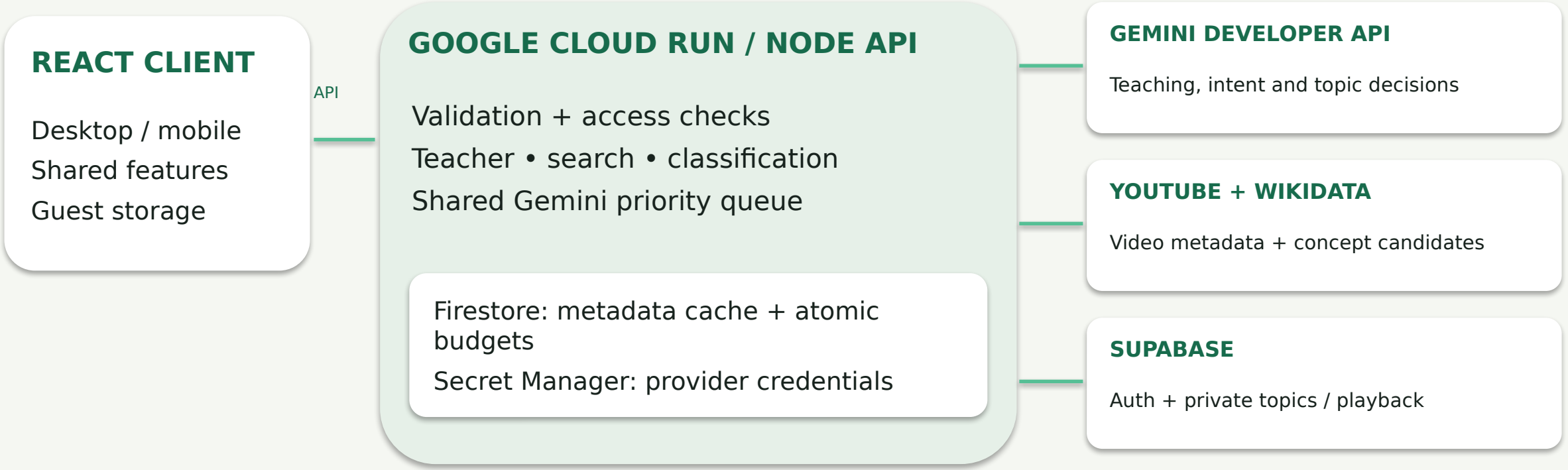
01 Clarify the learning angle before searching.



02 Keep the teacher beside the video.

Google Cloud runs the application. Gemini powers the learning decisions.

Latest repository architecture • browser calls the API; provider credentials stay on the server.



Video streams directly through the official YouTube player. No media is hosted by Revelink.

Three decisions that connect the experience.

TEACHER

Explain → act

Question + history + eligible video → structured answer, source basis and validated section or real video suggestions.

SEARCH

Intent → useful results

Question → learning choices or approval → YouTube retrieval → batched educational-purpose decisions.

CLASSIFICATION

Concept → discovery

Video metadata → Gemini topic extraction → Wikidata candidates → validated primary and related Universes.

Validated outputs / explicit uncertainty / cached decisions / teacher-first scheduling

A small, modular stack with clear responsibilities.

Experience	**React · TypeScript · Vite**	Shared learning features; dedicated desktop and mobile layouts.
Application	**Node.js · REST API**	Provider adapters, validation, background jobs and budgets.
AI & discovery	**Gemini API · YouTube Data API · Wikidata**	Teaching, educational intent, video retrieval and concept labels.
Cloud platform	**Cloud Run · Firestore · Secret Manager**	App hosting, persistent cache/counters and managed secrets.
Accounts	**Supabase Auth · PostgreSQL / RLS**	Sign-in, user-owned topics and playback records.
Verification	**Playwright · Node tests · Penpot**	Repeatable browser journeys, service rules and editable designs.

Designed for a small, quota-controlled pilot.

Illustrative compute estimate, not an actual bill or a quote for the Mumbai deployment.

REFERENCE COMPUTE

≈ $2.74

per month before free-tier credits

30 active hours · 1 vCPU · 0.5 GiB

30,000 requests

What the estimate includes

Reference request-based Cloud Run rates:

108,000 sec × ($0.000024 + 0.5 × $0.0000025)

+ 0.03 million requests × $0.40 = $2.739

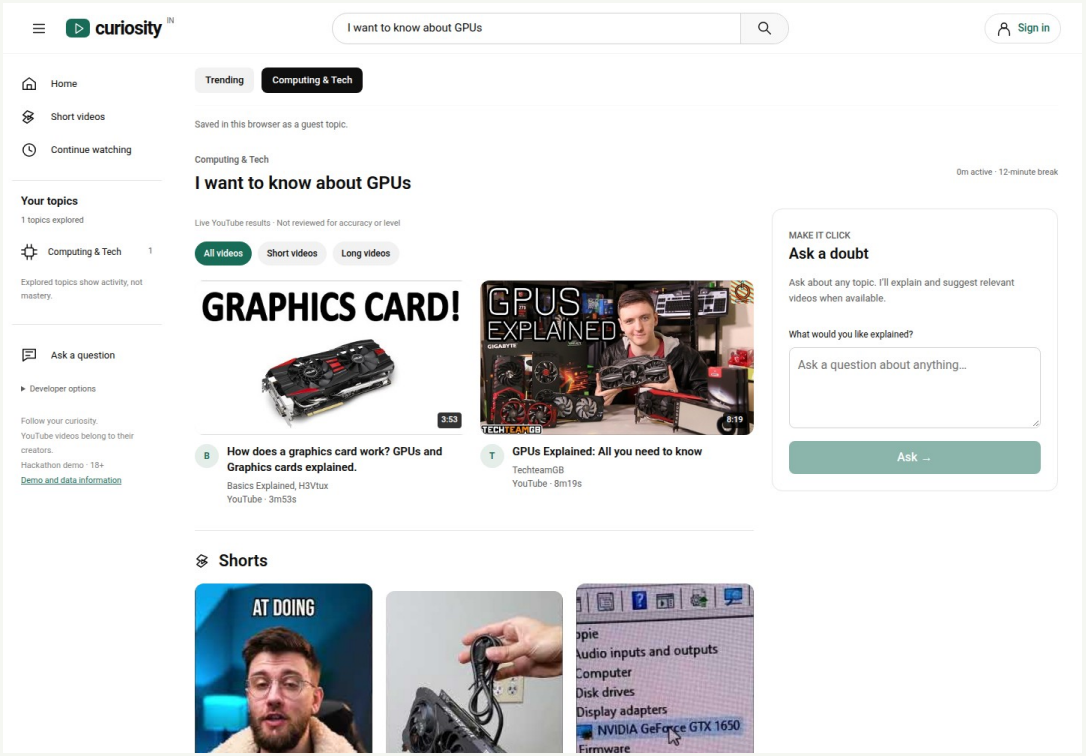
What must be priced separately

Gemini tokens / video input, Firestore, secrets, builds, image storage, network transfer, Supabase and labour. Region and billing tier change totals.

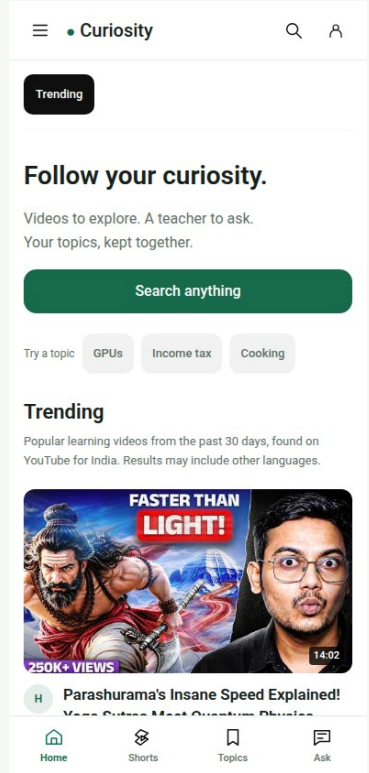
Rate reference: cloud.google.com/run/pricing • Full assumptions and exclusions are in the speaker notes.

A deployed product judges can open today.

Live Cloud Run screenshots • 18 October 2026 • deployed under the earlier Curiosity name.



Desktop: GPU results and the teacher workspace.



Same URL on a phone.

Evidence you can inspect. Limits we can name.

Fresh local verification + one live hosted teacher request + historical retrieval evidence.

95 / 95

Service / rule checks

10 experiment + 85 current tests

63 / 64

Browser journeys

Simulated APIs, accounts and player

PASS

Production build

Type checks + browser bundle

Live hosted check · 18 October

2.53 seconds: one hosted teacher response + two video suggestions.
Earlier YouTube check: 18 unique videos in 867 ms (13 October).

Single observations, not load tests. One legacy saved-search browser path failed; learning impact is unmeasured.

Prove the learning value. Then expand the map.

A focused next step, with outcomes measured before broader scale.

NEXT / RELEASE READINESS

Align the demo

Deploy the approved build, resolve the legacy-search regression, review real account sync and run a hosted end-to-end check.

NEXT / LEARNER PILOT

Test the core hypothesis

Recruit 10–15 adult learners. Compare time to resolve a doubt, task completion, context switches and voluntary return.

LATER / DISCOVERY

Connect adjacent ideas

Activate the reviewed learning graph, test progress assessments, and add distributed scheduling before multiple replicas.

SUBMISSION LINKS

Try one curiosity. Follow it through.

Search “GPUs” → ask a question → explore → return later.

FINAL PRODUCT

https://curiosity-u2ubbw6fra-el.a.run.app

PUBLIC GITHUB REPOSITORY

https://github.com/SubramDas/Curiosity

DEMO VIDEO · UNDER 3 MINUTES

[ADD PUBLIC DEMO VIDEO — UNDER 3 MINUTES]

SSRD / Subram Das / Watch. Discover. Connect the dots.